AKSHOBYA A S

Chikkamagaluru, Karnataka | +91 6366594660 | asakshobya@gmail.com

LinkedIn GitHub Portfolio

PROFILE

Information Science & Engineering undergraduate (2027 batch) with hands-on experience building full-stack web applications using the MERN stack, Django, and FastAPI. Built production-grade projects spanning AI-powered RAG applications with GPT-4o and Gemini, real-time system monitoring with ML-based anomaly detection, and analytics-driven PWAs. Proficient in Java, Python, and C with strong fundamentals in Data Structures, Algorithms, and OOP.

EDUCATION

Bachelor of Engineering (B.E.) – Information Science & Engineering **Expected 2027**
The National Institute of Engineering, Mysuru Current CGPA: 7.60

Pre-University (PCMC) **2023**
Creative PU College, Karkala Percentage: 96.33%

TECHNICAL SKILLS

- **Languages:** Java, Python, C, JavaScript, TypeScript
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js, Django, FastAPI, REST APIs
- **Databases & Vector Stores:** MySQL, MongoDB, SQLite, ChromaDB
- **Tools & DevOps:** Git, GitHub, Docker, Postman
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, System Architecture

PROJECTS

AI Semester Companion – AI-Powered Study Platform Code

- Engineered a full-stack AI study platform using **FastAPI** and **Next.js 15** with RAG-based document retrieval via **ChromaDB**, enabling context-aware chat over uploaded PDFs, DOCX, and PPTX files.
- Integrated **OpenAI GPT-4o** and **Google Gemini** APIs to auto-generate adaptive quizzes, flashcards, summary notes, and personalized study schedules.

Fuel & Service Tracker – Django PWA Live | Code

- Developed a full-stack Progressive Web App (PWA) using **Django** to track vehicle maintenance, fuel logs, and real-time mileage analytics with secure user authentication.
- Designed interactive **Chart.js** dashboards, offline support via service workers, CSV export capabilities, and achieved **WCAG 2.1 AA** accessibility standards.

ISHMS – Intelligent System Health Monitoring System Code

- Architected a real-time telemetry monitoring platform in **Django** collecting live CPU, memory, thermal, and power metrics via `psutil` exposed through a REST API.
- Implemented an ML anomaly detection pipeline utilizing **scikit-learn** (RandomForestClassifier) coupled with a rule-based root cause engine and live visualizations.

LANGUAGES

English | Kannada | Hindi